Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Data-Driven Digital Therapeutics Platform for Facial-Region Pain and Functional Limitations			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Bio・Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Information & Systems	Remote and Home Health care Devices	Therapeutic Devices & Diagnostic Devices	Bio-signal Measurement/ Diagnostic Device
Concept of Proposed Technology	The proposed technology is a **mobile-based digital medical device platform** designed to support **assessment, monitoring, and rehabilitation-oriented intervention** for **facial-region (orofacial) functional conditions** in at-home environments. The platform integrates: • Structured therapeutic modules for guided rehabilitation • Patient-data–driven personalization algorithms • AI-driven digital measures and longitudinal monitoring to enable objective functional assessment and real-world evidence generation Representative application areas include **facial-region pain–related functional limitations, neuromuscular dysfunction, temporomandibular joint disorders (TMD), and post-intervention rehabilitation**. Compared with limb musculoskeletal conditions,			

	facial-region functional disorders remain relatively underserved in scalable, home-based digital rehabilitation solutions. Accordingly, the proposed platform focuses on a **functional-domain–oriented scope** rather than a single disease label, enabling broader applicability while maintaining clinical relevance. The system is designed to enable **objective functional tracking, adaptive coaching, and longitudinal monitoring** in real-world home settings, supporting continuity of care beyond clinical visits through data-driven optimization of content, coaching intensity, and adherence support.
Rationale for Joint R&D Project	Joint R&D is required to translate an at-home, facial-region digital medical device concept into a **clinically credible, scalable, and quality-ready solution**. The facial-region functional domain presents unique biomechanical, neuromuscular, and behavioral complexity that necessitates close integration of clinical expertise with robust mobile engineering. Through joint R&D, the project enables: • Co-development of **standardized functional tasks**, clinically meaningful digital endpoints, and safety/escalation rules • Verification and validation of AI-driven digital measures and monitoring algorithms across heterogeneous mobile devices and real-world home environments • Integration of **patient-reported outcomes and longitudinal behavioral data** into personalization logic • A phased U.S. translation pathway, including (Year 1) market research and UI/UX localization research, (Year 2) usability/validation and reimbursement research, and (Year 3) go-to-market and reimbursement strategy development. Developing a reliable personalization AI layer requires shared access to clinically curated data, domain expertise for interpretation, and iterative verification and validation. These requirements cannot be fully addressed by a single organization alone. Given that facial-region digital rehabilitation tools are comparatively less mature than limb musculoskeletal solutions, **rigorous joint verification, validation, and feasibility evidence** are essential for downstream adoption and regulatory or quality readiness, including an evidence-oriented framework for real-world use (RWE) and guardrails for safe adaptation, and a staged plan to support U.S. adoption readiness.
Key Words	1. Orofacial function 2. Digital medical device 3. Personalization AI 4. Digital therapeutics

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3-4)	**End of the Project**	Level (6-7)
R&D Objectives	• Advance a mobile-based digital medical device platform for **assessment, monitoring, and rehabilitation-oriented intervention** in facial-region functional conditions • Develop a **patient-data–driven personalization layer** to adapt therapeutic content, coaching intensity, and adherence support over time • Implement AI-driven digital measures and algorithmic feedback to enable objective functional tracking and adaptive coaching in home environments, framing personalization			

R&D Target Specifications	as a decision policy that aims to maximize outcomes with minimal necessary intervention while reducing dropout risk under safety guardrails • Establish an evidence-oriented verification and validation (V&V) pathway, including pilot feasibility and measurement reliability evaluation for representative use cases and an evidence framework for real-world use • Support U.S. translation readiness through a phased workstream spanning (i) market research and UI/UX localization, (ii) usability/validation and reimbursement research, and (iii) go-to-market and reimbursement strategy development.
	• **Engagement / Adherence:** ≥ 70% weekly completion of prescribed therapeutic tasks in pilot evaluation • **Clinical / Functional Sensitivity:** Predefined improvement in a composite functional index and/or standardized patient-reported outcomes over the pilot period (final endpoints defined in protocol) • **Personalization Performance:** ≥ 10% relative improvement in adherence or functional score trajectory versus a non-personalized baseline (A/B or matched comparison where feasible) • Safety and Quality Assurance: ≥ 90% detection and flagging of invalid sessions or risk indicators with appropriate user guidance or escalation prompts (including clinically defined guardrails for adaptation) • Measurement Reliability (Supporting): ICC ≥ 0.80 for key digital-measure metrics under standardized repeat tasks (e.g., repeated functional tasks and patient-reported measures, as applicable)
R&D Plan	**1. Therapeutic Workflow Optimization** • Refine structured therapeutic modules, standardized assessments, and safety/escalation rules for facial-region functional conditions **2. Data and Personalization AI Development** • Define data schemas and develop adaptive algorithms for coaching, adherence support, and progress monitoring using longitudinal patient data **3. AI-Driven Digital Measures & Monitoring Engine Development** • Develop robust digital measures and monitoring algorithms with built-in data-quality assurance, reliability checks, and clinically interpretable scoring **4. System Integration** • Integrate therapeutic workflows, personalization logic, and digital-measure based feedback into a configurable, mobile-deployable platform **5. Verification, Validation, and Pilot Evaluation** • Execute V&V and pilot feasibility and reliability evaluation • Iterate system components and finalize an evidence package supporting downstream adoption and regulatory or quality readiness including an evidence-oriented framework for real-world use • In parallel, conduct U.S. readiness workstreams including usability testing, reimbursement pathway research, and preliminary go-to-market planning to support translation and adoption.

4. R&R (Roles and Responsibilities)

(max 10 pages)

Expected R&R of Research Institute	• Lead R&D planning and system design for an effective facial-region digital therapeutics platform • Define clinically meaningful therapeutic objectives, endpoints, and safety/escalation rules • Design V&V and pilot feasibility protocols, including standardized patient-reported assessments and functional task definitions • Provide clinical guidance for personalization logic, including risk stratification and response interpretation • Oversee data governance elements relevant to clinical validity, including labeling strategies, quality control, and bias considerations • Interpret pilot results and refine clinical requirements to support downstream evidence planning
Expected R&R of Korean SME Partner	• Implement and optimize the digital therapeutics workflow within a mobile-deployable product • Develop and integrate the patient-data–driven personalization AI layer • Build and integrate AI-driven digital measures and monitoring components with data-quality assurance and reliability checks • Deliver product-grade engineering, including privacy and security, QA/QC, cross-device testing, and deployment readiness • Prepare technical documentation and evidence packages to support downstream regulatory, quality, and commercialization planning

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Outcomes** • An integrated mobile-based digital medical device platform delivering structured digital therapeutics for facial-region functional conditions • A patient-data–driven personalization layer enabling adaptive therapy and adherence support • Clinically meaningful digital measures and objective functional monitoring with quality assurance, derived from multimodal patient inputs (patient-reported outcomes, behavioral/psychosocial surveys, and longitudinal usage data) • A concise V&V protocol and pilot feasibility evidence package addressing usability, engagement, safety, and measurement reliability • A U.S. translation readiness package including UI/UX localization recommendations, initial market landscape assessment, and reimbursement pathway research outputs. **Expected Impact** • Expansion of access to scalable, home-based therapy and monitoring for facial-region pain–related functional limitations and rehabilitation needs • Improved treatment engagement and continuity of care through personalized coaching and structured self-management • Establishment of a foundation for evidence-oriented adoption of facial-region digital therapeutics through clinically meaningful digital endpoints and validation outputs • Acceleration of real-world adoption by informing implementation, reimbursement, and go-to-market planning for the U.S. context.

(max 10 pages)

6. References

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